Advances in Ophthalmology and Optometry 2 (2017) 119–133

ADVANCES IN OPHTHALMOLOGY AND OPTOMETRY

# Evaluating Retinal Sequelae of Radioactive Episcleral Plaque Brachytherapy for Medium-Sized Choroidal Melanomas Using Optical Coherence Tomography Angiography

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**Keywords**

• Radiation retinopathy • Optical coherence tomography angiography (OCTA) • Plaque brachytherapy • Choroid • Melanoma • Retina • Tumor • Foveal avascular zone

**Key points**

- The Collaborative Ocular Melanoma Study validated episcleral plaque brachytherapy as a globe sparing alternative to enucleation for medium-sized choroidal melanomas.
- Innovations in the design of radioactive plaques have sought to minimize radiation complications, which compromise long-term vision and globe salvage.
- Vascular endothelial cell loss is thought to be the primary feature of radiation retinopathy.
- Different areas of the retina have demonstrated differential susceptibility to radiation, with the fovea being particularly radiosensitive.
- Optical coherence angiography can noninvasively assess the retinal vasculature and reveal subclinical anatomic changes that may precede overt clinical phenotypes leading to visual acuity decline in radiation retinopathy.

Disclosure Statement: The authors have nothing to disclose.

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http://dx.doi.org/10.1016/j.yaoo.2017.03.007
2452-1760/17/